CC-2029

FRI
Prosperity through research

Fie Research Institute

22/44, Ganganagar P. O., Near Sanjay Founders, ICHALKARANJI - 416116.
(Dist.Kolhapur), Maharashtra State, INDIA.
● Phone : (0230) 2441475 ● Cell : 98500 65882
● E-mail : fieresearch@gmail.com ● Website : www.fri-cal.com

CALIBRATION CERTIFICATE

ULR No.CC202923000012288F

OF FORCE MEASURING DEVICE

Date of Calibration	:	28/05/2023	Certificate No.	:	FRI/05/23/15005
Next Calibration due on	:	28/07/2025	Date of Issue	:	28/05/2023
Parameter	:	Force	Page No.	:	1/2

Calibrated for : M/s. Modtech Creative Labs Pvt. Ltd.,
Plot No.-3802, Laxman Vhar - II,
Gurgaon – 122 006, Haryana.

Customer Reference No. : Service Request Form Dated 25/05/2023

Details of instrument under calibration :

Instrument	: Proving Ring
Sr. No.	: 600 kN.060
ID No.	: PR/600 A
Capacity	: 600 kN
Type	: Integral
Dial Gauge Make	: Baker
1 Div.	: 0.002 mm
Dial Gauge No.	: TGI 113

Date of receipt : 28/05/2023

Mode of calibration : Compression

Calibration Method : The Proving Ring is calibrated in Compression mode as per FRI calibration Procedure No. FRICAL/CAL/01 based on IS: 4169-2014 / ISO: 376-2011.

Machine used for calibration : Dead Weight Type Hydraulic Multiplication Force Calibration Machine

Traceability : FRI/12/21/HFCM/037A* valid upto 23/12/2023

Dial Gauge Setting : Large pointer at 12 O' Clock position Small pointer at 05 revolutions.

Temperature : 24°C

Correction of temperature variation : Apply ±0.027% correction to each reading for each °C rise or fall of temperature.

Note :
1) Compression test were made out by using self-aligning Compression Pads Provided with the force - measuring device.
2) Prior to each reading, the dial gauge was lightly tapped on the centre of the dial cover.
3) This certificate refers only to the particular items submitted for calibration.
4) Creep is measured at zero load.

Calibrated by :	Checked by :	Authorised by :
Scientific Asst. (D.D.Magdum)	Scientific Asst. (U.V.Patil)	Sr. Engineer (R.V.Tambad)

CC-2029

FRi
Prosperity through research

Fie Research Institute

22/44, Ganganagar P. O., Near Sanjay Founders, ICHALKARANJI - 416116.
(Dist.Kolhapur), Maharashtra State, INDIA.
● Phone : (0230) 2441475 ● Cell : 98500 65882
● E-mail : fieresearch@gmail.com ● Website : www.fri-cal.com

CALIBRATION CERTIFICATE

ULR No.CC202923000012288F

OF FORCE MEASURING DEVICE

Date of Calibration	:	28/05/2023	Certificate No.	:	FRI/05/23/15005
Next Calibration due on	:	28/07/2025	Date of Issue	:	28/05/2023
Parameter	:	Force	Page No.	:	2/2

Details of instrument under calibration — Sr. No. : 600 kN.060
ID No. : Not Mentioned

Results — The calibration results are tabulated below :

Applied Force in kN	Deflection (Divisions)				Average (X_1,X_3,X_4) (Divisions)
	Unchanged Position		Changed Position		
	Series 1 at 0°		Series 2 at 120°	Series 3 at 240°	
	Increasing (X_1)	Increasing (X_2)	Increasing (X_3)	Increasing (X_4)	
100	167.2	167.2	167.4	167.4	167.3
150	251.4	251.4	251.6	251.6	251.5
200	337.0	337.0	337.2	337.2	337.1
250	421.4	421.4	421.6	421.6	421.5
300	506.2	506.2	506.4	506.4	506.3
350	591.8	591.8	592.0	592.0	591.9
400	677.8	677.8	678.0	678.0	677.9
450	764.4	764.4	764.6	764.6	764.5
500	850.4	850.4	850.6	850.6	850.5
600	1026.6	1026.6	1026.8	1026.8	1026.7
Creep at 30 sec.	0.4	0.6	0.4	0.6	
Creep at 300 sec.	0.2	0.2	0.2	0.2	

Classification: The Classification of force proving instrument is as follows:

Class	Mode	From	To	Uncertainty of measurement
Class 0.5	Compression	600 kN	150 kN	± 0.06%
Class 1	Compression	600 kN	100 kN	± 0.10%

The reported uncertainty is at coverage factor k=2 which corresponds to a Coverage probability of approximately 95% for a normal distribution, considering the relative error of different components such as Zero, Resolution, Repeatability, Interpolation and combining the uncertainty of Applied force.

Calibrated by :	Checked by :	Authorised by :
Scientific Asst. (D.D.Magdum)	Scientific Asst. (U.V.Patil)	Sr. Engineer (R.V.Tambad)